

Enhancing Software Evolution Reliability through Neurosymbolic Reasoning

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Abstract

Modern software systems are deeply embedded in interconnected dependency networks and must evolve continuously to remain viable. While Large Language Models (LLMs) have emerged as powerful tools for automating software evolution tasks, their probabilistic nature introduces fundamental reliability concerns: inevitable hallucinations and a lack of genuine logical reasoning make fully AI-driven evolution insufficient for industrial-grade demands. This thesis addresses this challenge by advocating a neurosymbolic approach that anchors the generative intelligence of LLMs within formal logical frameworks, systematically enhancing reliability across the three core phases of software evolution: environment perception, logic implementation, and artifact delivery.

For environment perception, we construct CompSuite, a reproducible dataset comprising 123 real-world Java client-library pairs that manifest incompatibility issues upon library upgrades. Sourced from 88 clients and 104 libraries, and accompanied by an automated command-line interface for single-command reproduction, CompSuite provides a high-fidelity empirical foundation for AI systems to perceive and reason about the risks propagated through dependency chains during software evolution.

For artifact delivery, we develop MPChecker, an automated technique for detecting multi-parameter code-documentation inconsistencies. MPChecker employs symbolic execution to extract inter-parameter constraints from code and leverages LLMs to parse corresponding constraints from documentation. A customized fuzzy constraint satisfaction framework reconciles the inherent uncertainty of LLM outputs and enables logical inconsistency detection. Evaluated on four widely used Python data science libraries, MPChecker discovered 14 real-world inconsistencies, of which 11 have been confirmed by the developers.

For logic implementation, we first identify and formalize the *Keyword Shortcut* bias in current code localization research, where models rely on superficial lexical cues rather than structural reasoning. We introduce KA-LogicQuery, a diagnostic benchmark for Keyword-Agnostic Logical Code Localization, which reveals catastrophic performance drops in state-of-the-art approaches. To address this, we propose DataLoc, a neurosymbolic agent framework that extracts program facts from codebases and synthesizes Datalog queries via an LLM, offloading structural traversal to a high-performance deterministic inference engine. Through parser-gated validation and mutation-based diagnostic feedback, DataLoc achieves precise, verifiable, repository-level code localization in a fully automated closed-loop workflow, significantly outperforming existing methods on logical localization tasks while maintaining competitive performance on standard benchmarks.

Collectively, this thesis demonstrates that reliable software evolution in the age of AI requires a synthesis of neural flexibility and symbolic rigor. By embedding neurosymbolic reasoning into each phase of the evolution lifecycle, this work provides both theoretical insights and practical tools toward building autonomous systems that are intelligent, verifiable, and trustworthy.